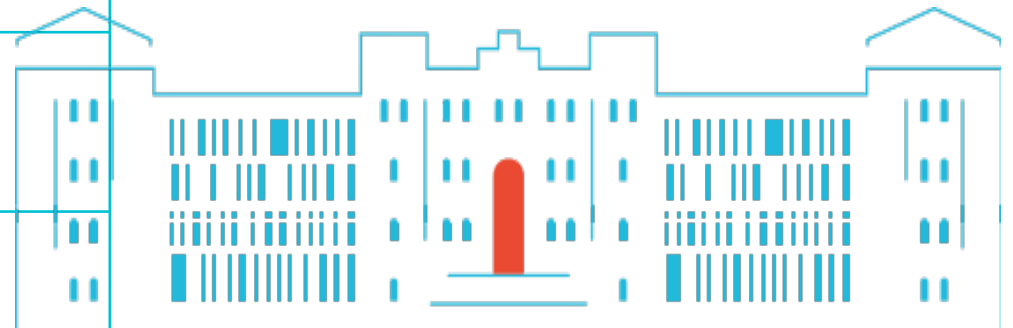
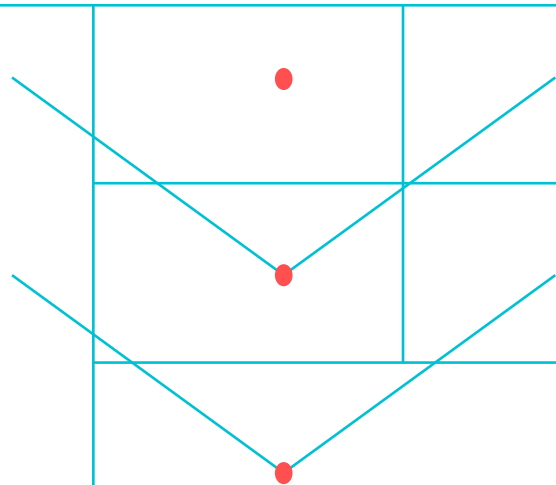


Numerical Methods for ODEs

How to submit Python programming
tasks via the online examination
system YAPS

TUHH
Technische
Universität
Hamburg

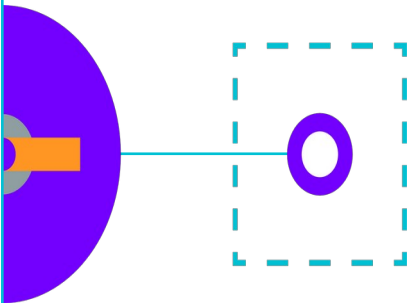


04.03.2026

Florian Bünger

Agenda:

1. How to log in and start
2. How to enter and run Python code
3. How to submit and finish



How to log in and start

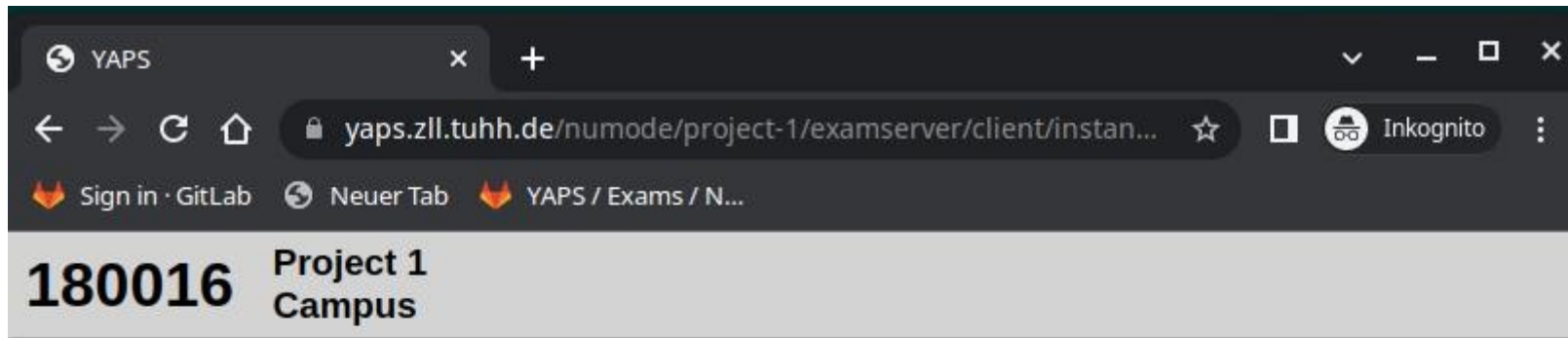
The YAPS link for submitting your code for Project 1 is at the bottom of this slide. Please, read the next lines carefully:

- Be sure that you have thoroughly tested your code offline and that it is ready for final submission via YAPS!
- You must use Chrome or Chromium as Webbrowser!
- Once you are logged in, you have 6 hours time to submit your Python program. In general this should take you only a few minutes so that there is no rush.
- After you submitted your code and finished YAPS you cannot reenter anymore at a later time.
- **Please, submit your code in the period 27.04. – 05.05.**

YAPS link for Project 1:

<https://yaps.amh.tuhh.de/numode/project-1-ss26/examserver/client/instance/>

How to log in and start



Numerical Methods for ODEs

Project 1 summer semester 2023
Examiner: Prof. Dr. Daniel Ruprecht

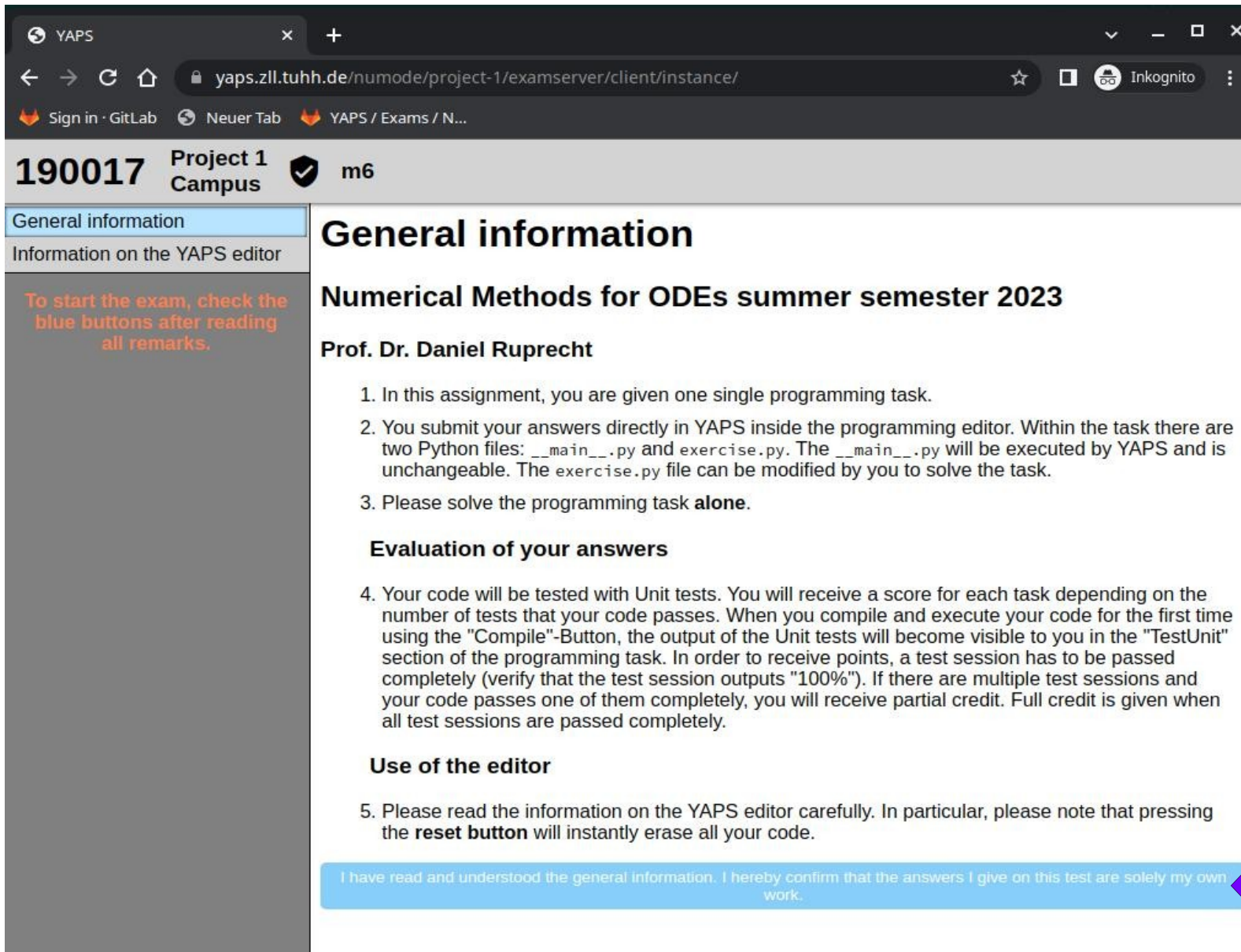
Log in with your university account.

By logging in, you agree to the temporary storage of your name as well as your IP address, email, user ID and matriculation number. Please also note the privacy policy of the Hamburg University of Technology at <https://www.tuhh.de/tuhh/ueber-uns/datenschutz.html>

A login form with two input fields: 'Account' and 'Password'. Below the fields is a green button with a white checkmark icon and the text 'LogIn'. A purple arrow points from the text 'Enter your TUHH-Account.' to the 'Account' input field.

Enter your
TUHH-Account.

How to log in and start: General Information



The screenshot shows a web browser window with the URL `yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/`. The page header includes the YAPS logo, a tab count, and navigation icons. Below the header, there's a section for "Project 1 Campus" with a checkmark and "m6". The main content area is titled "General information" and "Numerical Methods for ODEs summer semester 2023". It lists instructions for the exam, including reading remarks, solving the task alone, and using the editor. A blue button at the bottom is labeled "I have read and understood the general information. I hereby confirm that the answers I give on this test are solely my own work."

YAPS

Sign in · GitLab · Neuer Tab · YAPS / Exams / N...

190017 Project 1 Campus m6

General information

Information on the YAPS editor

To start the exam, check the blue buttons after reading all remarks.

General information

Numerical Methods for ODEs summer semester 2023

Prof. Dr. Daniel Ruprecht

1. In this assignment, you are given one single programming task.
2. You submit your answers directly in YAPS inside the programming editor. Within the task there are two Python files: `__main__.py` and `exercise.py`. The `__main__.py` will be executed by YAPS and is unchangeable. The `exercise.py` file can be modified by you to solve the task.
3. Please solve the programming task **alone**.

Evaluation of your answers

4. Your code will be tested with Unit tests. You will receive a score for each task depending on the number of tests that your code passes. When you compile and execute your code for the first time using the "Compile"-Button, the output of the Unit tests will become visible to you in the "TestUnit" section of the programming task. In order to receive points, a test session has to be passed completely (verify that the test session outputs "100%"). If there are multiple test sessions and your code passes one of them completely, you will receive partial credit. Full credit is given when all test sessions are passed completely.

Use of the editor

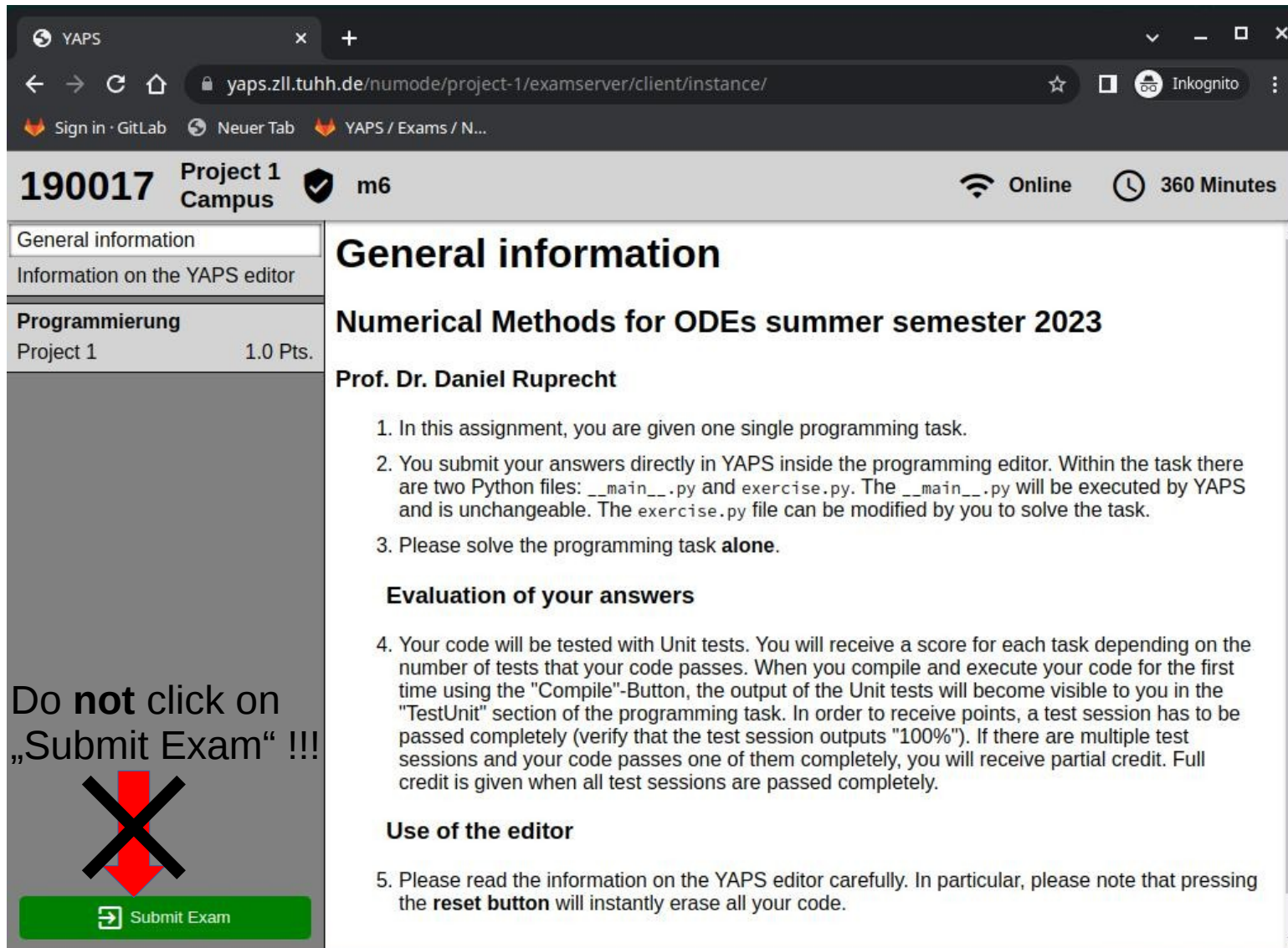
5. Please read the information on the YAPS editor carefully. In particular, please note that pressing the **reset button** will instantly erase all your code.

I have read and understood the general information. I hereby confirm that the answers I give on this test are solely my own work.

← Read the information.

← Click here to start.

How to log in and start: General Information



The screenshot shows a web browser window with the URL `yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/`. The browser tabs include 'YAPS', 'Sign in · GitLab', 'Neuer Tab', and 'YAPS / Exams / N...'. The page header displays '190017 Project 1 Campus' with a checkmark icon, 'm6', and 'Online' status with a 360-minute timer. The left sidebar contains a 'General information' tab and a 'Programmierung' section for 'Project 1' worth '1.0 Pts.'. The main content area is titled 'General information' and 'Numerical Methods for ODEs summer semester 2023' by 'Prof. Dr. Daniel Ruprecht'. It lists three instructions: 1. Solve one programming task. 2. Submit answers in YAPS using two Python files: `__main__.py` (unchangeable) and `exercise.py` (modifiable). 3. Solve the task **alone**. Below this is the 'Evaluation of your answers' section, stating that code is tested with Unit tests and that a 'TestUnit' section will show results. A final instruction, 'Use of the editor', notes that the 'reset button' erases all code. At the bottom left, a green 'Submit Exam' button is overlaid with a large red 'X' and the text 'Do not click on „Submit Exam“ !!!'.

YAPS

yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/

Sign in · GitLab Neuer Tab YAPS / Exams / N...

190017 Project 1 Campus m6 Online 360 Minutes

General information

Information on the YAPS editor

Programmierung

Project 1 1.0 Pts.

General information

Numerical Methods for ODEs summer semester 2023

Prof. Dr. Daniel Ruprecht

1. In this assignment, you are given one single programming task.
2. You submit your answers directly in YAPS inside the programming editor. Within the task there are two Python files: `__main__.py` and `exercise.py`. The `__main__.py` will be executed by YAPS and is unchangeable. The `exercise.py` file can be modified by you to solve the task.
3. Please solve the programming task **alone**.

Evaluation of your answers

4. Your code will be tested with Unit tests. You will receive a score for each task depending on the number of tests that your code passes. When you compile and execute your code for the first time using the "Compile"-Button, the output of the Unit tests will become visible to you in the "TestUnit" section of the programming task. In order to receive points, a test session has to be passed completely (verify that the test session outputs "100%"). If there are multiple test sessions and your code passes one of them completely, you will receive partial credit. Full credit is given when all test sessions are passed completely.

Use of the editor

5. Please read the information on the YAPS editor carefully. In particular, please note that pressing the **reset button** will instantly erase all your code.

Do not click on „Submit Exam“ !!!

Submit Exam

How to log in and start: YAPS editor information

YAPS

yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/

Sign in · GitLab · Neuer Tab · YAPS / Exams / N...

190017 Project 1 Campus m6 Online 358 Minutes

General information

Information on the YAPS editor

Programming Project 1 1.0 Pts.

This is a programming assignment

You can work directly in the YAPS editor and compile your code here.

Information on the editor

Info __main__.py exercise.py __main__backup.txt Compile & Run Reset

1

Diese Aufgabe als bearbeitet markieren? ☐

Zur nächsten Aufgabe

Editor

- **i** Information: This is where the task is.
- **E** Editable files: Change these files so that the code produces the output that is described in the task. Sometimes, you can check your code with a plot.
- **L** Not editable files: You cannot change these files, but they are accessible to the Python interpreter.
- **C** Compile: You can run your code here. The terminal output and potential images will appear next to the editor. You can compile your code once every 15 seconds.
- **R** Reset: This deletes all your changes and gets you back to the initial file content. **This will erase all your code without further confirmation needed!**

The following keyboard shortcuts are available:

- **Strg + c** Copy
- **Strg + x** Cut
- **Strg + v** Paste
- **Strg + f** Find and replace
- **Strg + z** Undo
- **Strg + y** Redo

Submit Exam

Read the information.

How to enter and run Python code

The screenshot shows the YAPS web interface. The browser address bar displays `yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/`. The page header shows '190017 Project 1 Campus' and 'm6' with a status of 'Online' and a timer at '357 Minutes'. The left sidebar has a 'Programmierung' section with 'Project 1' selected, showing '1.0 Pts.'. The main content area is titled 'Project 1' and '1 Punkt'. It contains a task description: 'Write a function `zombie(strategy, pa, pb)` that returns a time grid vector t of length $n = 101$ for an integration period of one year, i.e., $t[0] = 0$ and $t[100] = 1$, and a corresponding matrix x with n rows and three columns such that $x[i,0]$ is the number of humans, $x[i,1]$ is the number of zombies, and $x[i,2]$ is the number of removed zombies at time $t[i]$ for $i = 0, \dots, n - 1$. The parameter `strategy` stands for a mitigation strategy to encounter the zombie apocalypse. The following values are allowed: 0 : no intervention, 1 : training and arming humans, 2 : vaccination of humans. The parameters `pa, pb` $\in [-0.25, 0.25]$ represent the respective uncertainties in the model parameters α (efficiency of humans killing zombies) and β (efficiency of zombies infecting humans). Generate the results by using a suitable ODE solver in Python.' At the bottom of the task area are buttons for 'Info', '__main__.py', 'exercise.py', 'Compile & Run', and 'Reset'. A 'Submit Exam' button is at the bottom left. A toggle switch for 'Diese Aufgabe als bearbeitet markieren?' and a button for 'Zur nächsten Aufgabe' are at the bottom right.

Click here
... and here

The programming task is known from StudIP.

How to enter and run Python code: File `__main__.py`

The screenshot shows the YAPS web interface for Project 1. The browser address bar is `yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/`. The page header shows '190017 Project 1 Campus m6' and 'Online 354 Minutes'. The left sidebar contains 'General information' and 'Information on the YAPS editor'. The main area is titled 'Project 1' and shows a code editor for `__main__.py`. A purple arrow points to the file tab with the text 'Click here' and 'The file `__main__.py` cannot be changed. It will later run your code.' The code in the editor is as follows:

```
import numpy as np
import matplotlib.pyplot as plt
from exercise import zombie

def zplot(t,x,title,fname):
    path = "/home/student/result/"
    fs = 15 # FontSize
    Tend = t[len(t)-1];
    plt.figure()
    plt.rcParams.update({'font.size': fs})
    plt.plot(t, x[:,0], 'b', t, x[:,1], 'r', t, x[:,2], 'k')
    plt.legend(['Humans', 'Zombies', 'Removed'], loc = 'best')
    plt.xlabel('Time t')
    plt.ylabel('Number')
    plt.title(title)
    plt.xlim(0.0, Tend)
    plt.savefig(path + fname)
    N = len(t)-1;
    print('\n', title, '\n\t t = ', t[N], '\n\t #Humans = {:.2f}'.format(x[N,0]), '\n\t #Zom
    return

p = 0.25; # uncertainty of parameters alpha and beta in percent

t, x = zombie(0,0,0)
zplot(t,x,"Baseline scenario without intervention","pic_1.png")

t, x = zombie(1,0,0)
zplot(t,x,"Baseline scenario + Strategy 1","pic_2.png")

t, x = zombie(2,0,0)
zplot(t,x,"Baseline scenario + Strategy 2","pic_3.png")

t, x = zombie(1,-p,p)
zplot(t,x,"Worst case scenario + Strategy 1","pic_4.png")

t, x = zombie(2,-p,p)
zplot(t,x,"Worst case scenario + Strategy 2","pic_5.png")
```

At the bottom of the interface, there is a 'Submit Exam' button, a toggle for 'Diese Aufgabe als bearbeitet markieren?', and a 'Zur nächsten Aufgabe' button.

How to enter and run Python code: File exercise.py

The screenshot shows the YAPS web interface. At the top, the browser address bar shows 'yaps.zll.tuhh.de/numode/project-1/examserver/client/instance/'. Below the browser, the YAPS interface has a header with '190017 Project 1 Campus' and 'm6'. On the left, a sidebar shows 'General information' and 'Programmierung' with 'Project 1' and '1.0 Pts.'. The main area is titled 'Project 1' and shows a code editor for 'exercise.py'. The code is a Python script for a zombie model. At the bottom left, there is a green 'Submit Exam' button. At the bottom right, there is a button 'Zur nächsten Aufgabe'.

(1) Click here
The file
exercise.py can
be edited.

(2) Enter your code
here!

```
1 import numpy as np
2 from scipy.integrate import odeint
3
4 def zombie(strategy, pa, pb):
5     """
6     This function computes the evolution of the populations of humans and zombies
7     according to a chosen mitigation strategy.
8
9     Parameters
10     -----
11     strategy : integer
12         mitigation strategy
13         0 : no intervention
14         1 : training and arming humans
15         2 : vaccination of humans
16     pa : real number
17         uncertainty in the parameter alpha (= efficiency of humans killing zombies)
18     pb : real number
19         uncertainty in the parameter beta (= efficiency of zombies infecting humans)
20
21     Returns
22     -----
23     t : numpy.array
24         vector containing the time grid of the integration period of 1 year, t = [0,...,1]
25     x : numpy.array
26         x[i,0] : number of humans at time t[i]
27         x[i,1] : number of zombies at time t[i]
28         x[i,2] : number of removed zombies at time t[i]
29
30     Example
31     -----
32     >>> t, x = zombie(0,0,0) # Baseline scenario without intervention and with given \
33     >>> t, x = zombie(1,-0.25,0.25) # Strategy 1 in worst case scenario where alpha is 25% de
34
35     """
36
37     #####
38     #edit the code here#
39     #####
40
41
42     return ##
43
```

1 Punkt

Compile & Run **Reset**

Press **Reset**
if you want to get
back the initial
status of the
template.

(3)
Finally, press
Compile & Run
to run
__main__.py
which calls your
code.

How submit and finish

The screenshot shows the YAPS exam interface. On the left, a sidebar contains 'General information' and 'Information on the YAPS editor'. The main area is divided into three sections: 'Project 1' (showing code for 'exercise.py'), 'TestUnits' (showing test results), and a plot titled 'Baseline scenario without intervention'.

Project 1 Code:

```

1 import numpy as np
2 from scipy.integrate import odeint
3
4 def zombie(strategy,pa,pb):
5     """
6     This function computes the evolution of the populations of humans and zombies
7     according to a chosen mitigation strategy.
8
9     Parameters
10    -----
11    strategy : integer
12        0 : no intervention
13        1 : training and arming humans
14        2 : vaccination of humans
15    pa : real number
16        uncertainty in the parameter alpha (= efficiency of humans killing zombies)
17    pb : real number
18        uncertainty in the parameter beta (= efficiency of zombies infecting humans)
19
20    Returns
21    -----
22    t : numpy.array
23        vector containing the time grid of the integration period of 1 year, t = [0,
24    x : numpy.array
25        x[i,0] : number of humans at time t[i]
26        x[i,1] : number of zombies at time t[i]
27        x[i,2] : number of removed zombies at time t[i]
28
29    Example
30    -----
31    >>> t, x = zombie(0,0,0) # Baseline scenario without intervention and with
32    >>> t, x = zombie(1,-0.25,0.25) # Strategy 1 in worst case scenario where alpha
33    """
34
35
36
37
38
39
40

```

TestUnits:

```

x Schließen 'ed = 846.65
TestUnits
===== test session starts =====
platform linux -- Python 3.8.10, pytest-7.2.2, pluggy-1.0.0
rootdir: /home/student
collected 1 item
test/test_zombie.py .
===== 1 passed in 0.45s =====
pic_1.png

```

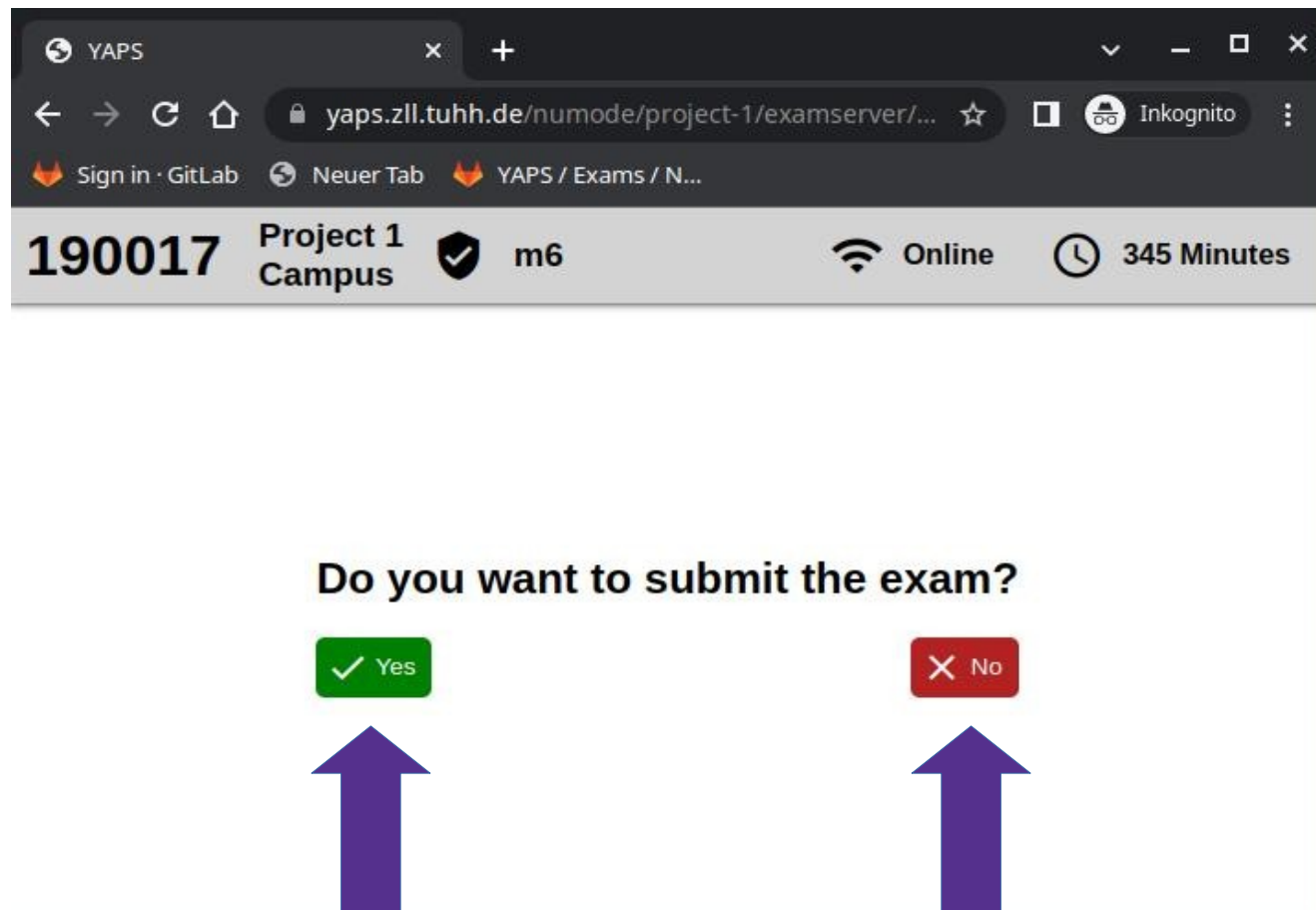
Plot: Baseline scenario without intervention

The plot shows the number of Humans (blue line), Zombies (red line), and Removed (black line) over time t (0.0 to 1.0). The y-axis ranges from 0 to 1000. Humans start at 1000 and drop to 0. Zombies start at 0 and rise to approximately 180. Removed starts at 0 and rises to approximately 820.

Annotations:

- (1)** If your code was correct and passed the internal test, then **== 1 passed ==** occurs. In this case you can submit and finish the exam, see (2).
- (2)** Press **Submit Exam** only if you want to finish.
- If your code was correct, then also some plots occur.

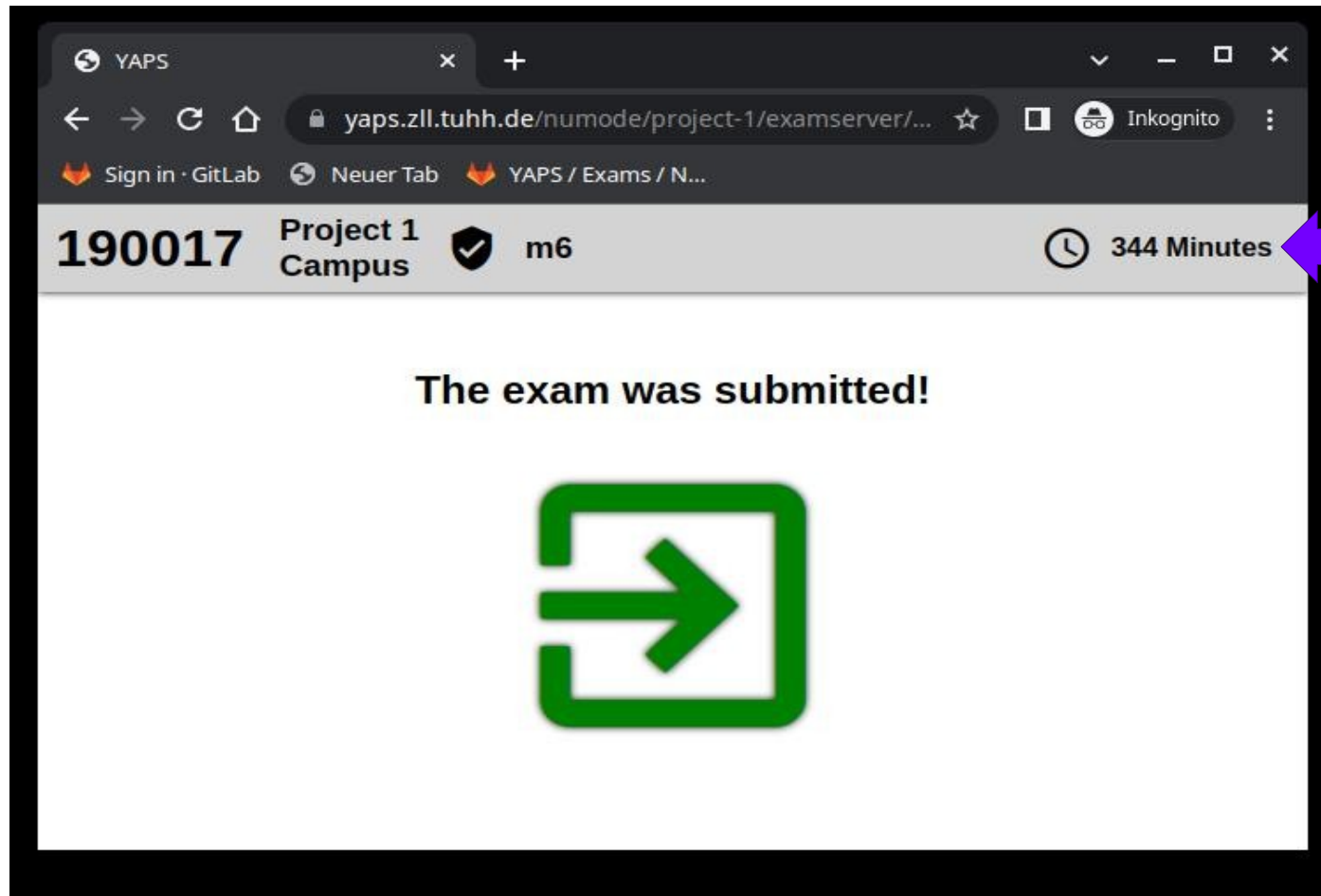
How to submit and finish



Click here to confirm that you really want to finish.

Click here, if you want to return to the YAPS editor.

How to submit and finish: Final confirmation



Remaining time that you did not use.